

MATERIALS PROCESSING (MT20202)

3-0-0

Offered by:

Metallurgical & Materials Engineering Dept.

Instructors:

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Important things for this course to be successful

- **Phone in flight mode/silent/off**



- **Attend every class**

- **Participate**

- **Do not be late**

- **Please do not talk among yourself.**

Marks

- Total:50
- Midterm-30
- Assignment/project- 20

Topics to be covered in the first half of the course

- Introduction, Materials
- Solidification (melting/casting)
- Welding/joining
- 3D Printing
- Thin-film Processing
- **Powder metallurgy**
- **Advanced Materials Processing (Nanomaterials)**

Introduction

- Why Materials Processing

What are different types of Materials?

Classification of materials

- Three basic types – based on chemical makeup and atomic bonding
 - METALS
 - CERAMICS
 - POLYMERS
- Combination of two or more of three basic type of materials
 - COMPOSITES
- Materials used in high-technology applications
 - ADVANCED MATERIALS (Nano-materials, electronic materials)

What type of materials are these?



Metals

Metals – High end applications



Aerospace and Defence



Space



Mining

Periodic Table of Elements: The Metals

I A												O					
1	2											10	11				
H	II A											III A	IV A	V A	VIA	VII A	He
3	4											5	6	7	8	9	10
Li	Be											B	C	N	O	F	Ne
11	12	III B	IV B	V B	VI B	VII B	VIII			I B	II B	13	14	15	16	17	18
Na	Mg											Al	Si	P	S	Cl	Ar
19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
55	56	57	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
87	88	89	104	105	106												
Fr	Ra	Ac	Rf	Db	Sg												

58	59	60	61	62	63	64	65	66	67	68	69	70	71
Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
90	91	92	93	94	95	96	97	98	99	100	101	102	103
Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lw

Metals & Alloys

- Composed of ? (Metallic (Al/ Fe/ Cu/ Ti/ Ni) and often also some non-metallic elements (C/ O/ N)) in relatively small amounts
- **Alloys:** Substance composed of ≥ 2 elements
- Metallic Bond: Valence electrons are detached from atoms, and spread in an 'electron sea' that "glues" the ions together.
- Characteristics:
 - Orderly arrangement of atoms: relatively dense
 - Stiff & Strong. Ductile, Resistant to fracture
 - Large number of non-localized electrons: High thermal & electrical conductivity
 - Opaque, reflective, shiny if polished.
 - Fe, Co, Ni have desirable magnetic properties

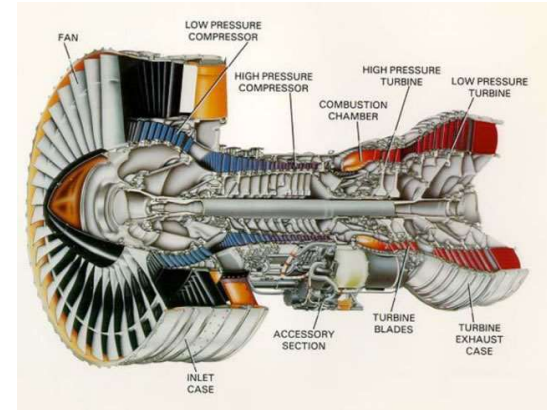
High end applications



Bridge



Mining



Aerospace and Defence



Space



Automobile

What type of materials are these?



Ceramics

Periodic table: ceramic compounds are a combination of one or more metallic elements (in light color) with one or more nonmetallic elements (in dark color).

I A																O					
1 H	II A															III A	IV A	V A	VIA	VIIA	2 He
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne				
11 Na	12 Mg	III B	IV B	V B	VI B	VII B	VIII					I B	II B	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar		
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr				
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe				
55 Cs	56 Ba	57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn				
87 Fr	88 Ra	89 Ac	104 Rf	105 Db	106 Sg																

Ceramics

- Compounds between metallic and nonmetallic elements: Oxides, Nitrides, Carbides
- Common ceramic materials: SiO_2 , Al_2O_3 , SiC , Si_3N_4 ,
Traditional ceramics: Glass, Clay (porcelain), Cement
- Ionic Bonding
- Characteristics:
 - Stiff & Strong like metals, Very Hard
 - Extremely brittle, highly susceptible to fracture
 - Insulators to heat and electricity. Resistant to high temperature and harsh environment
 - Opaque/ Translucent/ Transparent
 - Some oxide ceramics, like, Fe_3O_4 exhibit magnetic behavior

What type of materials are these?



Polymers

Periodic table with the elements associated with commercial polymers in color

I A																		O
1 H	II A											III A		IV A	V A	VIA	VIIA	2 He
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne	
11 Na	12 Mg	III B	IV B	V B	VI B	VII B	VIII				I B	II B	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr	
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe	
55 Cs	56 Ba	57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn	
87 Fr	88 Ra	89 Ac	104 Rf	105 Db	106 Sg													
			58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu		
			90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lw		

Naturally occurring vs. Synthetic polymer

There are two types of polymers: **natural** and **synthetic**

- Natural polymers occur in nature and can be extracted.
- They are often water-based.
- Examples: silk, wool, DNA, cellulose and proteins.
- Made by scientists and engineers.
- Synthetic polymers are derived from petroleum oil,
- Examples : nylon, polyethylene, polyester, Teflon, epoxy, synthetic rubber, PVC, silicone etc.

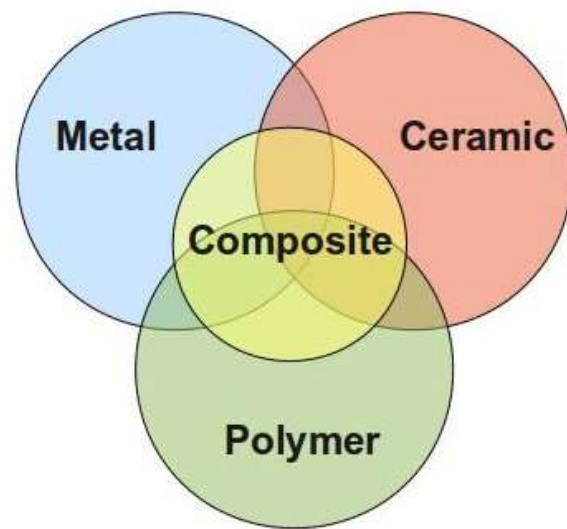
Polymers

- Plastics, Rubber, Organic compound based on C, H, O, N, Si
- Common polymeric materials: PE, PVC, Nylon, PC, PS, Silicone rubber
- Covalent Bonding: Large molecular structure, Chain like in nature with backbone of carbon atoms. Hydro-carbons abundant
- Characteristics:
 - Low density
 - Low stiffness and strength (<metals/ ceramics). However, specific strength may be high.
 - Extremely ductile → good formability to complex shapes
 - Insulators to heat and electricity
 - Nonmagnetic
 - Chemically inert & unreactive to many environments
 - Soften or decompose at modest temperatures

Thoughts about these “fundamental” Materials

- **Metals:**
 - Strong, ductile
 - high thermal & electrical conductivity
 - opaque, reflective.
- **Ceramics:** ionic bonding (refractory) – compounds of metallic & non-metallic elements (oxides, carbides, nitrides, sulfides)
 - Brittle, glassy, elastic
 - non-conducting (insulators)
- **Polymers/plastics:** Covalent bonding → sharing of e's
 - Soft, ductile, low strength, low density
 - thermal & electrical insulators
 - Optically translucent or transparent.

Composite Materials



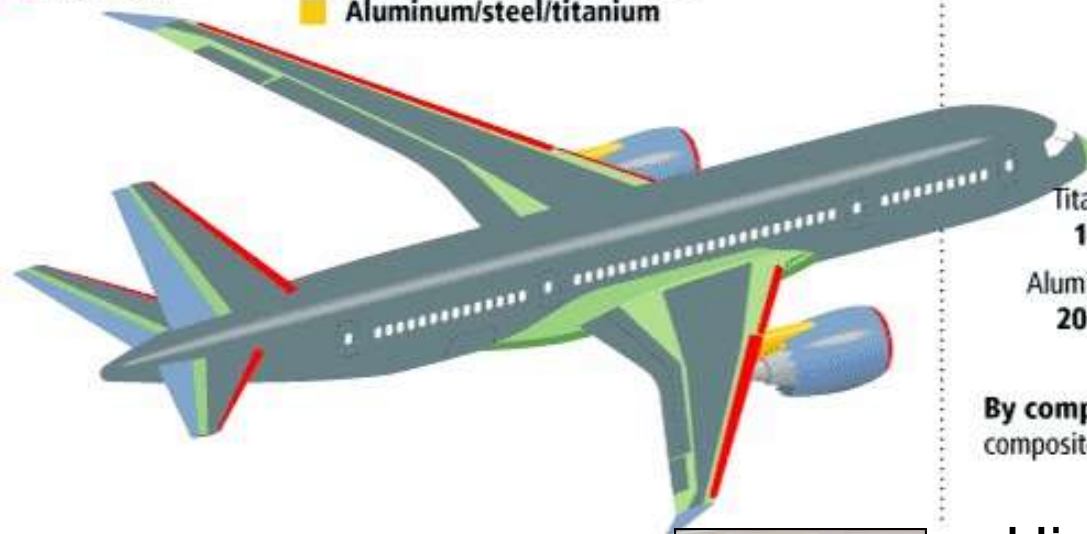
Combination of good of both the worlds:

Composite Materials

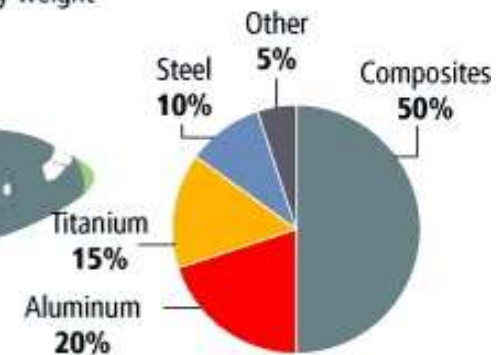
- Composed of 2 or more individual materials
- Naturally occurring:
 - **Wood** (Cellulose fibers + matrix of lignin),
 - **Bone** (Crystallized mineral platelets (nm thick) + Collagen as matrix)
- Man made:
 - Fiberglass (Glass fibers (strong, stiff but brittle) embedded in polymeric material (ductile, flexible, less dense but weak)) (**GFRC**) ,
 - Carbon fiber reinforced polymer composite (**CFRC**)
 - Used in **aerospace application, bicycles, golf-club, tennis rackets, ski/ snowboards.**

Composite materials

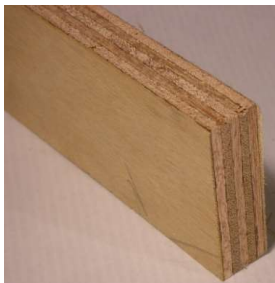
Materials used in 787 body



Total materials used
By weight



By comparison, the 777 uses 12 percent composites and 50 percent aluminum.



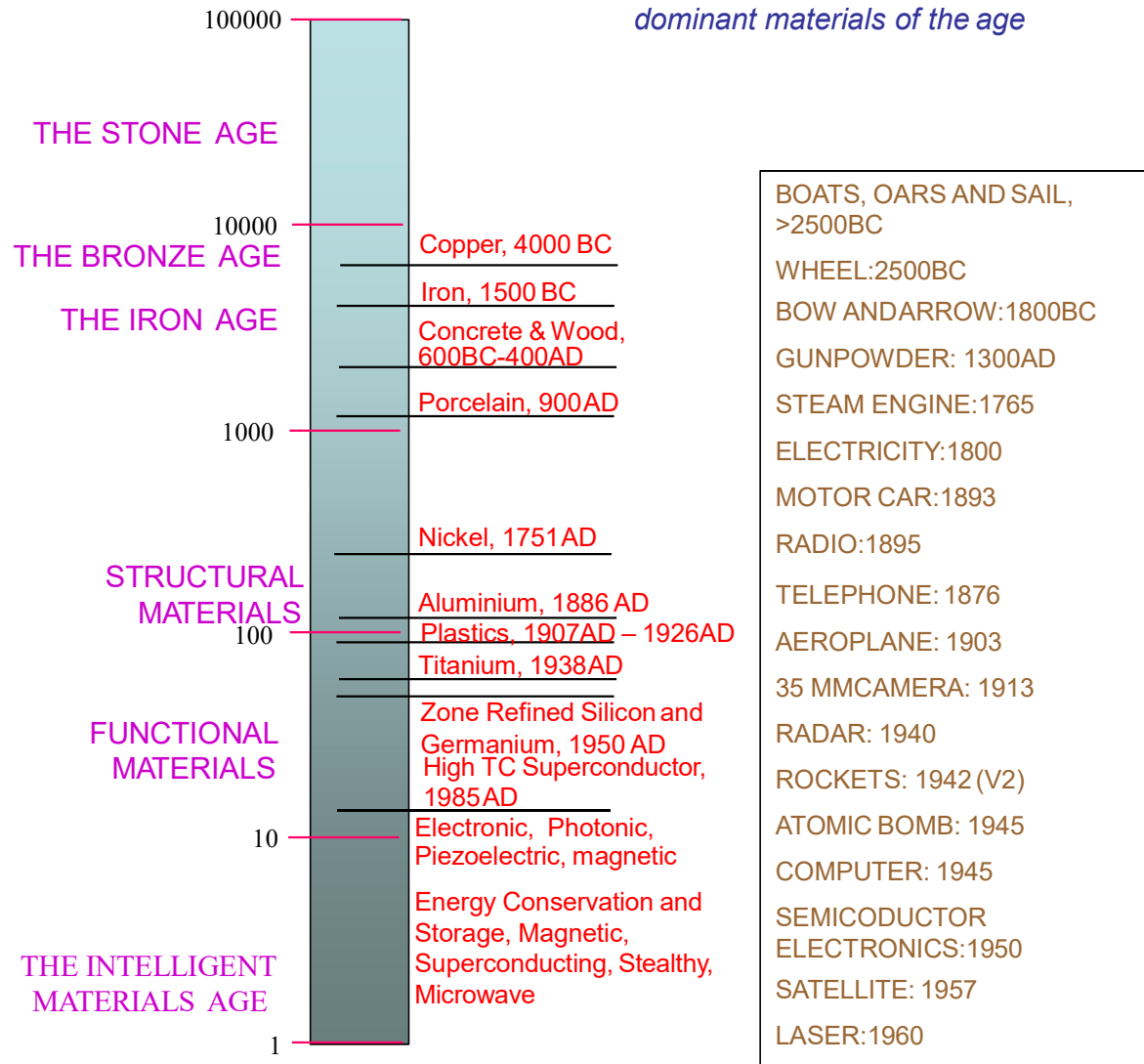
High Tech sporting

- Golf club
- Tennis rackets
- Skis/ Snowboards

History of materials processing

A HISTORICAL EVOLUTION

The profound influence of materials on human civilization is recognised in the identification of distinct time periods with the dominant materials of the age



THE FIRST GREAT INNOVATION: METAL FORMING

NATIVE COPPER HAD TO BE HEATED TO ABOUT 800C TO SOFTEN,
REWORK AND HARDEN

Neolithic Period (Late
Stone Age)

8000 - 4000
BC

Chalcolithic Period (Copper
Age)

4000 - 3150
BC

THE EARLIEST OBJECTS OF COPPER
WERE FOUND IN IRAN AND ANATOLIA



THE SECOND GREAT INNOVATION: A METAL FROM ORE

A SMALL PIT DUG IN THE GROUND, WITH A LOW SUBSTRUCTURE OF FIELD STONES VENTILATED BY GOATSKIN BELLOWS

IRON ORE WAS ADDED AS FLUX TO COPPER ORE AND CHARCOAL

THE METALLIC COPPER PRODUCED BY THIS PROCESS REMAINED AT THE BOTTOM OF THE FURNACE AS AN IRREGULAR INGOT

5000-3000 BCE:

PROBABLY THE EARLIEST COPPER INGOT IN HISTORY TIMNA, ISRAEL

"A land whose stones are iron and out of whose hills you can dig copper."
Deuteronomy 8:9

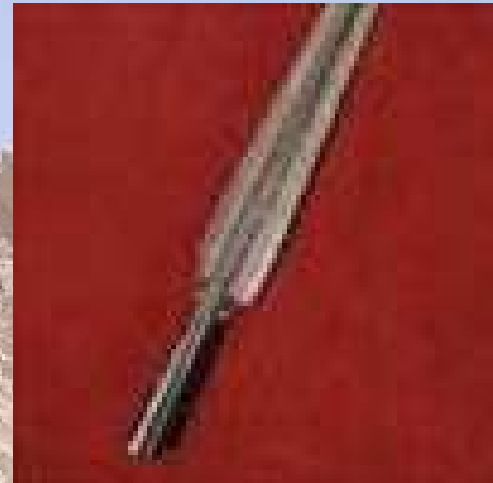


THE THIRD GREAT INNOVATION: ALLOYING

TIN STRENGTHENS CAST COPPER

Early Bronze Age	3150 - 2300 BC
Middle Bronze Age	2200 - 1550 BC
Late Bronze Age	1550 - 1200 BC

THE EARLIEST BRONZES
HAVE BEEN FOUND IN
EGYPT



THE EXPANSION OF THE MATERIALS WORLD



James Stuart, Professor of
Engineering at Cambridge
1875 - 1890

- **In his day:** a few hundred materials
 - *No polymers* – now over 45,000
 - *No light alloys* – now several thousand
 - *No composites* – now hundreds
- **Today:** > 160,000 engineering materials???? **Actually an infinite variety**
- **Creates two needs:**
 - 1Education** – how best teach materials?
 - 2Industry** – how manage materials information ?